

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	560631
Award Program	Tipping Points
Project Title	Integrating Community and Modeling Efforts to Evaluate the Impacts and Tradeoffs of Food System Interventions
Grant Start Date	02/01/2018
Grant End Date	12/31/2022
Total Grant Budget	\$2,100,718
Total FFAR Budget	\$1,100,000
Matching Funder(s)	\$1,000,718
Final Report Date	05/01/2023
Project Director/Principal Investigator Information	
Full Name	Becca B.R. Jablonski
Email	Becca.jablonski@colostate.edu
Phone	970-491-6133
Grantee Organization Information	
Organization Name	Colorado State University
Address	2002 Campus Delivery
City, State Zip	Fort Collins, CO 80523-2002
Tax ID	846000545
Authorized Signing Official (ASO) Information	
Name	Catherine Douras



ASO Title	Senior Research Administrator
Email	Catherine.douras@colostate.edu
Phone	970-491-2375

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Colorado, all 8 congressional districts

- 1.2. How many new jobs were created because of this grant funding?

1

- 1.3. How many jobs were maintained because of this grant funding?

0

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

In 2017, Colorado State University (CSU) and the City and County of Denver received \$1M from FFAR's inaugural round of the Tipping Points program. The goal of the grant was to provide Denver with data and a model to better understand tradeoffs associated with their proposed municipal food procurement policies (in other words, how to prioritize food purchase decisions and incentives). The project leveraged >\$5.9M in on-going food system programming throughout the state of Colorado, and built off recommendations from the 2017 Denver Food Vision (facilitated by the Mayor-appointed Denver Sustainable Food Policy Council and signed by the Mayor), the Colorado Blueprint for Food and Agriculture (facilitated by the Governor-appointed Food Systems Advisory Council), and ongoing work of the Colorado Food Policy Network (an umbrella network for local food policy councils, task forces, coalitions, and farm to school networks across Colorado). When the COVID-19 pandemic hit, FFAR provided an additional \$100,000 in funding to integrate new data collection and research into the project scope.

To our knowledge, this is the first research project in the U.S. to conduct a rigorous, multi-disciplinary evaluation of tradeoffs associated with various scenarios of implementation of the Good Food Purchasing Program (GFPP). GFPP is perhaps the most widely adopted municipal food procurement strategy in the U.S., including 25 cities and \$1.1 billion in annual food expenditures as of 2022.

Primary results from the applied research project include:

- 1) Farmers make decisions that are not profit-maximizing for many reasons (Hale et al. 2020; Toulabi et al. 2022). Thus, understanding the socio-cultural attributes associated with producer decision-making is key to modeling realistic supply chain responses.
- 2) One size does not fit all when looking at the impacts of various third-party certification programs on farmers, communities, and the environment. Accordingly, there needs to be careful consideration as to whether or not a particular verification program will achieve local goals (Jablonski et al. 2020).
- 3) Facilitating dialog between urban food policymakers and rural producers to understand potential tensions, mitigate such tensions, and capitalize on opportunities is essential. Our work supported new opportunities for urban-rural dialogue (see [here](#) and [here](#)), which changed the way that Denver is implementing GFPP (Jablonski et al. 2022).
- 4) Policymakers and advocates often assume widespread benefits of school meals, but there is little evidence to support some of these assertions. In our research using national data, we did not find strong evidence supporting beneficial spillovers of school meals on a household's quality of food-at-home acquisitions (Cleary et al. 2021).
- 5) Though we found some misalignment in terms of the size of Denver's municipal procurement, what the school system can purchase, and current agricultural production in Colorado, we found evidence of tradeoffs across key environmental (CO₂ and water) and economic indicators (school costs, share of costs to producers or supply chain businesses, producer profitability, producer variability in profitability) for different procurement scenarios being considered under the GFPP scheme.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture

and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

Perhaps our most important and implementable finding is to include stakeholders that will be affected by municipal procurement policies in decision-making and implementation processes and to work to build bridges and trust across diverse actors. The work conducted as part of our FFAR grant is transferable and actionable in other cities across the U.S., and can be encouraged through U.S. Department of Agriculture (USDA) funds and strategy.

Despite significant Federal funding to support local and regional food systems, cities are increasing leading the charge in developing local food market opportunities. For example, Johns Hopkins University Food Policy Networks project now includes 322 food policy councils, most of which are in urban areas, and many of which are implementing value-based food procurement policies.

Prior to this project, in 2015, the City and County of Denver began developing its first long-term strategic vision for the future of food in the city and county. The Denver City Council and the Mayor's Office supported the development of a comprehensive, citywide food plan to bring about greater alignment across the wide range of stakeholders with a clearer understanding of the role of government in food systems planning and development. The resulting document, the Denver Food Vision and Plan, emerged from a public engagement process throughout the city and county (but not the region). One area of emphasis included in the Vision's 'winnable goals' was that 25% of all food purchased by public institutions in the City and County of Denver come from Colorado sources by 2030. Subsequently, a mayoral advisory committee was established (part of the Denver Sustainable Food Policy Council) to propose implementation recommendations. Unfortunately, despite the fact that Denver's food production cannot meet the 25% goal, rural and regional stakeholders were not included in those conversations or given power/agency.

Leveraging the FFAR funds, the City and County of Denver hired its first Institutional Food Procurement Administrator (a position now made permanent by the City). The Administrator was tasked with building a broader coalition to support Denver's food procurement policy and implementation strategy, so that impacts of different scenarios could be more widely discussed and understood. The Coalition has now been meeting for several years; it provides a space for different viewpoints to be represented while educating members about policy related topics such as agricultural labor and animal welfare issues. Meetings now regularly include participants from hospitals, universities, schools, farm organizations, labor, and the city of Denver, among others. As a result of hiring the Administrator and coordinating coalition meetings, the CSU research team had a

straightforward way to disseminate data and model results and get feedback. Further, the coalition discussions have resulted in changes to how Denver is implementing its good food purchasing procurement strategy (for example, differentiating across types of livestock production systems and focusing less on organic certification in dryland production systems). An example of an unintended consequence of the Coalition is that Denver Public Schools (DPS) and the Colorado Potato Administrative Committee (potatoes are the largest growing fruit or vegetable commodity grown in the state) applied for a joint grant to explore opportunities to get Colorado grown potatoes into the schools.

Another replicable strategy tested as part of the FFAR grant was to establish an annual Colorado Food Summit (coloradofoodsummit.com). The Summit is a collaborative effort, led by CSU, the Colorado Department of Agriculture, the Colorado Food Systems Advisory Council, and all 12 of the state's local food policy councils and coalitions. The Summit inspires diverse Coloradan food and agricultural practitioners to build shared understandings of the challenges and opportunities facing our food system. In 2022, the event hosted 480 participants from Colorado (it sold out over 1 month in advance), representing all parts of the food system, including 40 youth and FFA Colorado State Officers. Complete evaluation results from the 2022 Food Summit can be found here:

<https://foodsystems.colostate.edu/events/foodsummit2022/evaluation/> Most participants reported gaining greater knowledge about the food system, including such topics as food and agricultural policy, markets, production practices, and diverse stakeholder values and beliefs. Over 85% of survey respondents expressed that they were able to learn about the firsthand experiences of someone with a different perspective during the event. More than half (56%) of those surveyed reported finding common ground with others at the event. Perhaps most significantly, 2,125 new connections were made during the summit and more than 90% of participants were planning to follow up on connections made during the event!

Through several new programs that that USDA has recently launched – especially the Regional Food System Partnership program (<https://www.ams.usda.gov/services/grants/rfsp>) and the new Regional Business Centers program (<https://www.ams.usda.gov/services/local-regional/rfbcpr>) there are now financial resources that USDA could tie to these types of intentional, regional strategy.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

Food systems play a critical role in achieving multiple United Nations Sustainable Development Goals (UN SDGs). Accordingly, many city, national, and multinational governments are actively implementing sustainable food policies, with “local” institutional procurement programs being the most common. Local food procurement programs reach about 368 million children globally, and are supported by a total investment of \$70B U.S. annually. Governments are creating policies to leverage this spending in order to achieve multiple SDGs, including reducing hunger and promoting good health and well-being (UN SDGs 2 and 3), creating opportunities for decent work and economic growth (UN SDG 8), and creating or growing markets for sustainably produced food (UN SDG 12). In the United States (U.S.), there is a dedicated Federal funding program to support farm to school programs, which helped to spur the reported \$1.26 billion spent on locally procured items by schools in the 2018-19 school year. Additionally, 43 states, Washington D.C. and the U.S. Virgin Islands successfully passed 170 bills and 70 resolutions supporting farm to school efforts between 2002 and 2020. The GFPP is perhaps the most widely adopted municipal food procurement strategy in the U.S., including 25 cities and \$1.1 billion in annual food expenditures as of 2022.

“Local” or values-based school procurement policies involve tradeoffs that are often not fully considered by government officials, planners, and advocates. To our knowledge, this is the first research project in the U.S. to conduct a rigorous multi-disciplinary evaluation of tradeoffs associated with various scenarios of implementation of food procurement policies.

There are five primary findings of our research:

1) Farmers make decisions that are not profit-maximizing for many reasons. Thus, understanding the socio-cultural attributes associated with producer decision-making is key to modeling realistic supply chain responses (Relevant project publications from this line of work: Hale et al. 2020; Toulabi et al. 2022). Understanding producer responses to different market-based interventions are also key to understanding if and how a proposed effort will be successful. Integrating producers early on into urban food policy discussions can thus be key if the success of the intervention requires a particular type of producer response (e.g., crop shifting, certification, change of rotation, investment in soil health). Some cities are starting to hire regional planners to work on topics such as food and agriculture. These point people can play key intermediary roles. For example, New York City has regional planners who, among

other tasks, support a NYC Regional Food Policy Working Group. Integrating Colorado producers into Denver food purchasing decision discussions via the Good Food Purchasing Program Coalition has – and will continue to have – important impacts.

2) One size does not fit all when looking at the impacts of various third-party certification programs on farmers, communities, and the environment. Accordingly, there needs to be careful consideration as to whether or not a particular verification program will achieve local goals. (Relevant project publication from this line of work: Jablonski et al. 2020). This means that prior to implementing a food procurement policy, cities (and their regional partners) should work to establish clear goals. These could be focused on keeping a certain amount of land in agriculture, improving profitability of socially-disadvantaged producers, increasing supply chain resilience, etc. Once those goals are established, identifying which strategy(ies) are most likely to achieve those goals is key. Partnerships with University researchers who can help to bring data and models to bear can help inform decision-makers. But, ultimately tradeoffs will exist across implementation pathways.

3) Facilitating dialog between urban food policymakers and rural producers to understand potential tensions, mitigate such tensions, and capitalize on opportunities is essential. Our work supported new opportunities for urban-rural dialogue (see [here](#) and [here](#)), which changed the way that Denver is implementing GFPP. For example, under the GFPP municipal buyers get extra points for purchases of organically grown or raised products. Results from our research wherein DPS demand more bread made from organic flour (wheat) grown in Colorado demonstrate positive profitability impacts to producers that transition acreage to organic production (including small and midscale farms). However, results also show that in Colorado's dryland farming system (where wheat is primarily grown) there are environmental tradeoffs associated with shifts to additional organic acreage due to the inability to practice no-till, and thus increased erosion resulting in negative soil health outcomes. With this information, Denver's procurement strategy is placing less emphasis on organic wheat certification and more on production practices shown to both improve farm profitability and soil health (including supporting the Colorado Department of Agriculture on a Colorado-specific [voluntary certification program](#)).

(Relevant publication from this line of work: Jablonski et al. 2022).

4) Policymakers and advocates often assume widespread benefits of school meals, but there is little evidence to support some of these assertions. In our research using national data, we did not find strong evidence supporting beneficial spillovers of school meals on a household's quality of food-at-home acquisitions.

(Relevant project publication from this line of work: Cleary et al. 2021). One implication of the work – particularly in light of unprecedented amounts of state and Federal funding available to support school meals that prioritize local – is ensuring a strong evaluation partner. Designing an evaluation plan from the onset is important to understanding if an intervention results in the impacts that are proposed. In many cases, practitioners may assume positive effects that are not occurring. Rigorous evaluation of these programs and policies are critical to understand impact and effectiveness.

5) Though we found some misalignment in terms of the size of Denver's municipal procurement, what the school system can purchase, and current agricultural production in

Colorado, we found strong evidence of tradeoffs across key environmental (CO₂ and water) and economic indicators (school costs, share of costs to producers or supply chain businesses, producer profitability, producer variability in profitability) for different procurement scenarios being considered under the GFPP scheme. Thus, the ways in which good food purchasing is implemented results in different economic and environmental outcomes, which are important for decision-makers to consider depending on their sustainability goal(s). As with above, being clear on what the goal(s) are of an intervention at the onset can help decision-makers to make determinations about what tradeoffs are acceptable.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

This project included four primary objectives:

1) Integrate available primary and secondary data to depict the status of Denver's food system, its interactions with provisioning producers, and to characterize the behavior of four agent-types (producers, processors, retailers, and consumers).

2) Build an integrated, dynamic, computational model of the current food system that links the diet quality, food security, environmental and overall socio-economic condition components. For this project, we will integrate three distinct computer modeling and simulation techniques. The first, agent based modeling (ABM), will be used to represent the structure and behavior of the four agent-types and their representing indicators. The second, the Object Modeling System (OMS), integrates the agent-types as modular components via their indicators, data inputs feeds, and linkages described in objective 1, including integrating previously developed modeling components that have been developed as a modeling solution. The modeling solution handles "what-if" scenarios – such as those 'shocks' in the Denver Food Vision. The third, Cloud Services Integration Platform (CSIP), will allow the scalable deployment of our modeling solution as standardized web-services to be accessible and used from the project's outreach tool: communitycommons.org. All models used herein are open source, and code with appropriate metadata will be made available through application lifecycle management (ALM) so that the approach can be adapted to other problems and cases.

3) Develop targeted hypothetical intervention scenarios ('shocks') based on the winnable goals from the Denver Food Vision and the action items of the CO Blueprint of Food and Agriculture. Scenarios will be developed in conjunction with community partners such that the model is framed appropriately and trained to incorporate realistic reactions to system changes. All results will be validated through feedback exercises with the producer and community advisory committees.

4) Engage stakeholders in communicating findings and integrating models in decision-making. Stakeholder will better understand how to use the data and model in decision-making, why tradeoffs might exist among choices, and best practice interventions based on existing community resources that optimize realistic and measurable diet quality, food security, environmental and overall socio-economic conditions outcomes.

Very detailed explanation of what was accomplished under each of these objectives is explained in 2.6 of this report. To the best of our knowledge, we accomplished the intent of what we set out to do (though perhaps not in the exact way that it was originally specified in the original report).

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

There were a number of changes in the project, due to how our partners in Denver determined to implement their food procurement strategy, research findings that informed the model, and due to COVID.

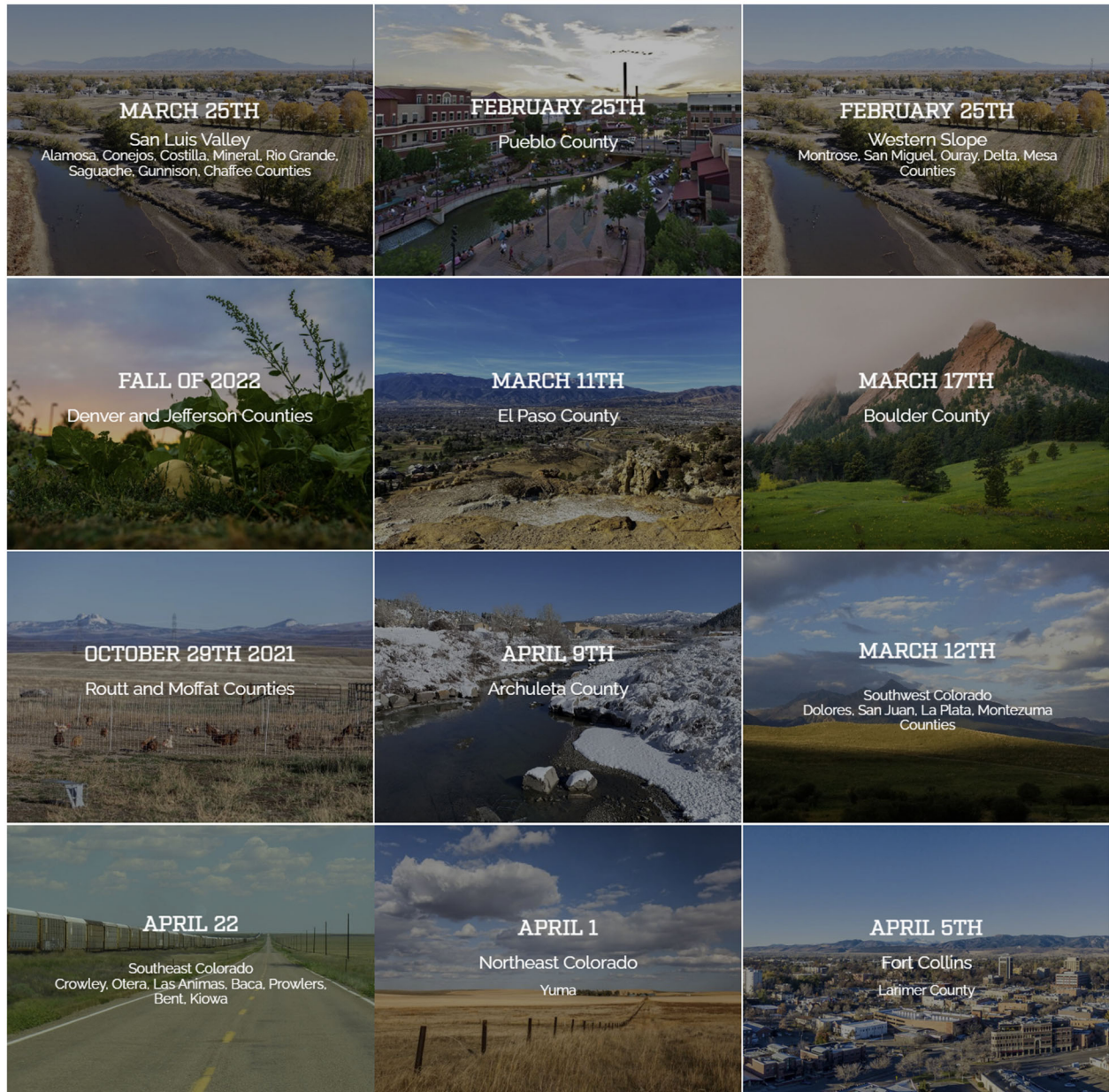
First, early into our project The City and County of Denver decided to focus their food procurement policies on the Good Food Purchasing Program: <https://goodfoodpurchasing.org/>. Thus, our team pivoted to using the GFPP pillars as our external model 'shocks'. Perhaps most significantly, the group focused on household data/models had to pretty significantly change its focus. Given that the GFPP will not directly impact what households purchase for food at home consumption, this research team had to refocus to understand how institutional purchases (in their case schools) will impact household dietary quality for households with kids that participate in the national school lunch program. This pivot to the Good Food Purchasing Program is very timely and relevant example to develop scenario analysis on given the number of cities that have adopted it, including Los Angeles, Chicago, Washington DC, New York, and Philadelphia.

Second, once this pivot was made, the research team found no evidence of spillovers from school meal acquisitions to the quality of households' food acquisitions overall (see publication by Cleary et al.). School meal acquisitions are generally healthier than other forms of food-away-from-home for households more than 130% of poverty thresholds. For both SNAP and SNAP-eligible-non-participating households, but the team did not find evidence of school meals acquisitions' spillover to the quality of households' food-at-home or food-away-from-home acquisitions. Accordingly, within the model, we do not see any impact to households based on the various food procurement scenarios.

Finally, due to COVID we had to rethink some of our some of our outreach and validation planning. We were unable to meet with some of our producer stakeholders in-person during the height of the pandemic, meaning that many conversations about data had to be virtual.



We were (unfortunately) forced to reimagine the 2021 CO Food Summit as it was not possible to bring together 400-500 stakeholders. Accordingly, we pushed the event date back to December 8, 2022 and planned 12 regional convenings across the state (learn more: <http://coloradofoodsummit.com/>). Each of these convenings was planned and executed by their local food policy council/coalition, and were pivotal to making the 2022 Colorado Food Summit such a success. By holding the regional convenings, we were able to ensure that the content for the statewide meeting emanated from the regions through a more inclusive process. Further, as part of this process, the planning committee (leads of each of the councils/coalitions) worked with an equity consultant to help the group better understand why there is distrust across the food system and how to serve as better leaders to build bridges in their communities and across divides.



- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this

section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

This project included four primary objectives:

1) Integrate available primary and secondary data to depict the status of Denver's food system...

Early into the project, based on conversations with the city and county of Denver, we pivoted the project a bit to more realistically reflect how Denver was planning to implement its food procurement strategy. After submitting the grant application to FFAR, Denver decided to recommend the Good Food Purchasing Program, and one of the largest buyers that signed on to the baseline assessment was the DPS. Given challenges in gathering data along the supply chain (given its proprietary nature), we ended up focusing on three pillars – farm level (everything pre-farm gate), supply chain level (farm-gate to DPS), and households (spillover impacts to households resulting from school meals). Further, we focused on four commodities that were of key importance both to DPS and agriculture within the state: potatoes, peaches, wheat, and beef.

We used a huge number of primary and secondary data sources, as well as existing models, within our model, and validated these data with producers, supply chain stakeholders, and secondary data (where we were leveraging models).

To provide some additional detail for potatoes and beef, data that we used are specific to counties in CO, U.S.A., with potatoes grown in the San Luis Valley in Saguache, Rio Grande, and Alamosa Counties and beef in Routt and Moffat Counties. Potato producers are distributed based on mapped areas of center-pivot irrigation using data from the Colorado Department of Natural Resources, which correspond with areas mapped in CropScape. Beef commodities are made spatially explicit through random placement in the appropriate counties. Agricultural statistics from the National Agricultural Statistics Service provided the number, acreage, and other noneconomic attributes for producers, and the U.S. Department of Agriculture's Agricultural Resource Management Survey data, 2013-2016, provided the financial attributes.

To reflect observed cropping practices or to support scenario analyses, russet potatoes that are traditionally produced at high volumes are represented as well as differentiated potatoes. This is because DPS could receive points under the environmental sustainability category of the GFPP for potatoes that are certified organic. We do not do the same for beef as the focus is on source-identified (from Colorado) for DPS to obtain the points. In the case of beef, DPS has incentives under GFPP to purchase CO raised products. As most beef processed at a large scale, multinational processing plant cannot be verified to have been raised in CO, we include grass fed beef from Northwest CO.

Physically based ecosystem process models allow us to understand environmental baseline and scenario conditions, so we created potential production estimates for a variety of climatic and management conditions for each of the commodities. Main production estimates for

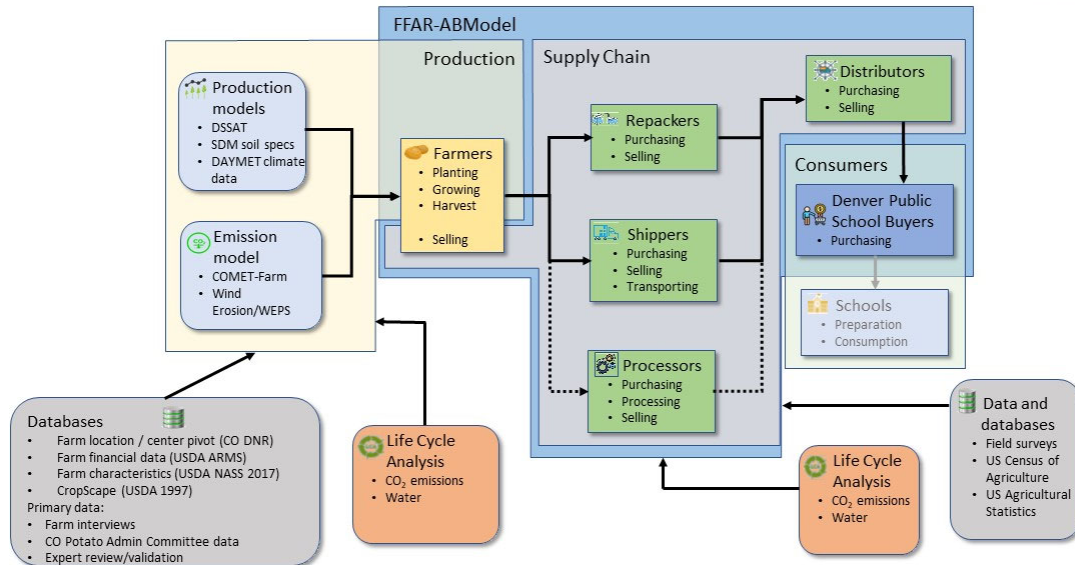
potatoes and wheat were from Decision Support System for Agrotechnology Transfer (DSSAT). DSSAT was integrated into the Cloud Service Integration Platform (CSIP), which offers a “Model-as-a-Service” platform and capabilities at scale. Model input scenarios were JSON service requests for modeling spatially placed parcels owned by producers with recommended management practices. CSIP included embedded simulation tools for calibration and a collection of data services accessing spatial and nonspatial databases. Here, a set of supporting CSIP data micro-services automatically provisioned the DSSAT service with climate data using Daymet and soil data from SSURGO adjusted with pedo-transfer functions when necessary.

CSU Extension Specialists estimated fertilizer regimes for DSSAT simulations in conventional and organic production. We generated DSSAT scenarios for 40 years of climatic conditions on 511 Daymet centroids for potatoes in San Luis Valley. We calibrated DSSAT cultivar parameters for potatoes (conventional/organic Canela Russet, conventional/organic Purple) and minimized the bias of estimated yield against U.S. Department of Agriculture Census reported averages. The resulting yields were stored in raster maps and used as crop production input for the agent-based model.

Given its extensive nature, beef production is modeled differently and had different data requirements. Beef producers are first created and parameterized using county-level data. Then, landscape surfaces of variables such as mean forage production, public-private lands, grazing lands (both public and private), and irrigated lands are loaded. Ranches are then distributed according to the landscape variables, such as expected land ownership per cow based on local forage production. The main dynamic input in the beef submodel is annual precipitation, based on a 30-year sample (1981-2010) of interpolated values.

Metrics tracking the effects of production on erosion and biogeochemical responses such as carbon storage are generated for the same locations and sets of conditions using Wind Erosion/WEPS and COMET-Farm, which integrates simulations from DAYCENT.

Colorado Potato Food System



2) Build an integrated, dynamic, computational model of the current food system...

We built an agent based model (ABM), which digests the data described above, providing estimates of changes in production system environmental indicators, including soil erosion potential, soil carbon storage, and nutrient and water use efficiency.

The ABM is implemented in NetLogo and is comprised of modules representing common processes related to the modeled commodities. At initialization, a spatial arena is scaled to the state of CO at 1 km² resolution. Commodity-specific agents such as potato farmers, repackers, shippers, and processors are created at this stage, as are general agents such as distributors and DPS.

Continuing the detailed example for potatoes, farmers are initialized with a farm size and landscape patches that may be planted to potatoes or barley, the typical rotational crops in the region. Assets and maximum inventories, an expected sell price (modified during simulation), and attributes of the farmer including gender, age group, race category, and farmer experience are set for each farmer. These farmer attributes are used to calculate an initial transition likelihood score (TLS) for each farm. Purchase prices and conventional and organic maximum inventories for Canela Russet and Purple Majesty potatoes are set for each repacker, shipper, and processor. These are set for the distributors and DPS as well. Scenarios may cause different sets of parameters to be initialized to values appropriate to each situation.

To track the creation and fulfillment of contracts, tables are created for each of the conventional and differentiated products in ABM using the 'Tables' extension of NetLogo. Tables are mutable data structures that allow for rapid searching using keys, which here are owners of commodities seeking supply chain members with which to enter into contracts. Each keyed entry in a table stores the identity of an agent for which a contract has been established and the quantities and prices of product contracted for and delivered to date. This structure allows for the efficient representation of sparse square matrices for each type of transaction without the computational challenges inherent in such matrices.

Event Scheduling

A simulation starts in week 30 of year 1 and continues for 10 years using a 1-week timestep, such that 52 steps represent one year. The calendar week is an important controller of events in this discrete-time simulation tool. For potatoes, those that remain in storage after local procurement has occurred each year in week 31 (late July) are sold to entities outside CO that are not represented explicitly in the model, thereby emptying inventories.

Producer Decision-making

Based on our interviews with producers as well as the peer-reviewed literature, producer behavior and response to policy are heterogeneous and must be accurately incorporated into food systems modeling to understand potential impacts of policies. Decisions regarding attributes to include for producers are based on a literature review and extensive interviews with diverse producers across our four focal commodities.

Expected profits for each farm using conventional and organic methods are calculated and compared. If conventional methods are more profitable, a large farm will remain in conventional farming; small and medium farms may still transition if their TLS, which spans 0 to 1, and is used to incorporate multiple sociocultural capital elements simultaneously in producer decision-making, is decremented. If organic methods are more profitable, selected 1-km² areas within large farms are assigned to begin transitioning, which takes three years. For small and medium farms, attributes of the owners are used to alter the TLS, such that first-time farmers are more likely to transition (TLS increased by 10%). A designation as an alternative farmer based on farmer attributes (i.e., gender, age, first-time farmer, and race category) is used to distinguish farms with owners apt to be more experimental in their management.

Our interviews and literature review suggest that conventional and alternative farmers respond differently to practices of their neighbors. Therefore, for each small or medium farm apt to be alternatively managed, up to three of the geographically nearest farms with the same attributes are examined to ask if they are transitioning, or have transitioned, to organic practices. If a neighboring farm has transitioned, the TLS increases (+10%). If two or more have done so, the score increases by a larger amount (+20%). By contrast, large and medium farms that are apt to be traditionally managed decrease by the same amounts if neighboring farms are transitioning or are organic.

Lastly, if yields for any differentiated potato product are <70% of average yield of Canela Russet potatoes, it is considered a year with below-normal production and the TLS is

incremented by 2.5%. Likewise, if yields for a particular potato type are <70% of that type's long-term average, the year is considered poor in production and TLS is increased 10%. These changes make farmers incrementally more likely to transition to organic farming in response to perceived poor production years, which is a finding from our qualitative fieldwork.

For farms with $TLS \geq 0.5$, the proportion of farmed land in organic production is calculated. If that value is above 25%, all farmed lands are marked for transitioning to organic. Otherwise, a random 1 km² of owned land in conventional production is selected and assigned as transitioning to organic. For farms with $TLS < 0.5$, if a farm is small or medium and managed by an alternative farmer and is fully organic, it remains so. All other farms below the TLS threshold will convert any areas in organic production back to conventional production. Lastly, 1-km² areas designated as in transition for more than 3 years are marked as certified organic.

Negotiate Contracts

Contracts account for power dynamics across the supply chain, and they dictate how each dollar of sales is distributed. Three contracting periods are used to sell potatoes each year, with one-half of expected product sold in late July (week 31) and one-half of the remaining inventory sold in each of two off-season periods: late January (week 5) and late April (week 18). In each of these weeks, past contracts are concluded and new contracts are established. Programmatically, the tables that store information about contracts are cleared. Then the sale of potatoes through farmers, shippers, repackers, and distributors is implemented.

Variations in the agents involved are scenario-dependent, such as a GFPP scenario in which DPS contracts directly with shippers rather than through the intermediary distributors to emulate a shortened farm-to-school supply chain. For each product type, the prices buyers are willing to offer are adjusted. From a pool of an adjustable number of buyers reflecting market knowledge, sellers select the most favorable price available to them and interact with that buyer. At the end of every contracting period, the buyer offering the highest price and buyer offering the lowest price each adjust their offering prices by 10% (downwards for the highest bidder and upwards for the lowest bidder). In this way, a market equilibrium emerges.

The capacities of buyers to store potatoes is based on their inventory. Sellers calculate how much to sell based upon their yields, inventory, and desired number of transactions, and then contracts are established. Buyers with the highest purchase prices precede those with lower purchase prices, with the seller selling as many potatoes to each buyer as the buyer's capacity and the seller's inventory allows for at the agreed upon price, until the seller has sold all their potatoes. Incomes for sellers and expenses for buyers are tracked. Table structures within the ABM store the identifiers of buyers a given seller has contracted with, total quantities to be provided, prices at which product is to be exchanged, and a running tally of product delivered. Analogously, the anticipated inventory spaces of buyers are decremented as contracts are put in place.

Production, Contract Fulfillment, and Consumption

In early September (week 36), potatoes are produced. Spatial surfaces reporting production estimates for the year simulated as generated from the ecosystem models are loaded into the ABM simulation. Mean yields are calculated for all lands planted in each commodity. In week 37 (mid-September), as well as weeks 6 (early February) and 19 (early May), contracts are fulfilled if there is sufficient production. For each seller, the quantity available to sell is compared to the promised quantity under a given contract, the lower amount is provided to

the buyer, monies are exchanged, and inventories are updated. The final purchaser in the supply chain, DPS, provides product to schools, who in turn provide meals to students each week, depleting DPS inventory.

To account for impacts to household health, we econometrically evaluated the relationship between the Healthy Eating Index (HEI) of households' food-at-home acquisitions and the number of meals acquired by children attending a National School Lunch Program participating school. Using U.S. Department of Agriculture FoodAPS data (collected during a period when schools transitioned to new nutrition standards), we found limited evidence of the positive contribution of school meals to the quality of households' food acquisitions overall. These findings are also supported by recent research using National Health and Nutrition Examination data. In light of our econometric findings, we used HEI changes to individual students documented in the literature. The model ends with one DPS purchasing agent, which buys all the school food the district would need. The DPS purchasing agent closes the contracting/financial exchange part of the model, which we then translate to changes in HEI based on the findings of Smith et al.

Life Cycle Assessment

Life cycle assessment methodology was used to incorporate key environmental outputs across supply chains into our ABM. Comparing environmental impact metrics of various products alongside financial data enables us to infer prices for units of different types of environmental impact. Particular attention is paid to agriculture and land use activities, as they account for 71% of food systems greenhouse gas emissions, and the food system is responsible for a third of global anthropogenic emissions.

The system boundaries for potatoes included all farming operations and inputs used at the farm during one growing season (i.e., from after harvest of the previous crop to harvest of the target crop) with a functional unit defined as 1 kg of produce at the farm gate. Emission factors for farming operations (machinery and fuel use), seeds, irrigation, and fertilizers, herbicides, and pesticides were retrieved from the Ecoinvent 3.6 database using cutoff modeling, which provides recycled material burden-free to subsequent processes. This is especially relevant for the application of manure in organic systems, which do not carry any of the emissions related to producing the manure. Direct N₂O emissions from fertilizer use were obtained from the DSSAT model. Activity data for farming operations was obtained from DSSAT and the Ecoinvent database. Greenhouse gas emissions were calculated with the Tool for Reduction and Assessment of Chemicals and other environmental Impacts (TRACI 2.1), which uses a 100-year global warming potential of 25 for methane and 298 for dinitrogen monoxide. Water consumption was obtained from the ReCiPe midpoint indicator using the hierarchical approach.

The system boundary for beef included cattle production, transportation to the processing plant, carcass processing to produce ground beef, and transportation from the plant to DPS. All ground beef was assumed to originate from cull cows. Therefore, farm-gate inventory and emissions data were mass allocated between the calf and cow based on the total annual live-weight sold from the operation. Farm-gate activity data were obtained from an earlier study of the environmental footprints of U.S. beef production. The operations modeled for that study were designed to be representative of regional management practices and environmental

characteristics, and were based on a published survey of management characteristics of western beef cattle production systems.

Activity data for processing in the conventional scenario were derived from an LCA of conventional beef production in the U.S. Primary activity data for beef processing for the small, locally-owned meat processor scenario were obtained from a small meat processor in northwest Colorado. The functional unit was 1 kg of ground beef delivered to DPS. As with the other commodities, greenhouse gas emissions were calculated with TRACI 2.1, and water consumption was obtained from the ReCiPe midpoint indicator using the hierarchical approach.

Outputs and Analyses

Information mapped for commodities produced in CO can be customized in the spatial arena (e.g., product, rotation, ownership, transition likelihood score). The interface reports quantities of commodities purchased by DPS and quantities of commodities sold to others outside the school system. The average inventory of producers and DPS are plotted. Indicators of the environmental impacts of production systems are reported and expected changes in CO₂ equivalents emitted and water use are reported. NetLogo's BehaviorSpace is used to conduct experiments that involve multiple simulations with different parameter settings. Based on the output, we can see dynamic changes and trade-offs that occur throughout the food system in economic, environment, health, and sociocultural domains.

3) Develop targeted hypothetical intervention scenarios ('shocks')....

Though it took us a bit longer than anticipated to develop the 'shocks' -these ended up being based on actual ways that Denver was considering implementation of the Good Food Purchasing Program. The shocks are specific for each commodity and were designed to understand economic, environmental and social changes that would take place if the City of County of Denver, and DPS (DPS) adopt the GFPP. For peaches: Baseline scenario: DPS does not purchase peaches in the baseline. Peaches go through wholesale and alternative market channels. CO Fresh scenario: DPS purchases fresh local peaches. Individually quick frozen (IQF) scenario: A portion of culled peaches are processed by an IQF facility. DPS purchases IQF peach product. For potatoes: Baseline scenario: DPS purchases fresh fries from distributor for all potato product needs, nothing from CO growers is purchased. CO Fresh scenario: DPS purchases conventional russets for all potato product needs. Good Food Purchasing Program (GFPP scenario): DPS purchases conventional russets directly from shippers. Small potato scenario: DPS purchases small conventional russet potatoes from shippers. Specialty product: DPS purchases a specialty product made from conventional (or organic) purple potatoes that are processed at a CO facility. For wheat: Baseline scenario: DPS purchases flour and sliced bread milled from conventional hard red winter wheat. Organic scenario: DPS purchases flour and sliced bread milled from organic hard red winter wheat. White whole wheat scenario: DPS purchases flour and sliced bread milled from conventional Snowmass wheat. For beef: Baseline scenario: DPS purchases conventional culled ground beef neither born, braised, nor processed in a specific state. American Grassfed Alliance (AGA) scenario: DPS purchases culled ground beef with AGA certification. Colorado source identified (CSI) scenario: DPS purchases culled ground beef with CSI certification. Animal Welfare Approved (AWA) scenario: DPS purchases culled ground beef with AWA certification.

4) Engage stakeholders....

Stakeholders were engaged throughout the project as explained elsewhere in the report. Most importantly, stakeholders developed the shocks, provided and validated data, worked with the research team to understand and validate supply chains, participated in data collection to understand decision-making, and were involved with results dissemination – in part through the GFPP Coalition.

The establishment of the Good Food Purchasing Program Coalition is well described other places so will not be elaborated upon here. In this section we will focus on other stakeholder engagement activities.

From the onset we worked hard to build trust with our target producer groups. There is a lack of trust in Colorado between many of these agricultural commodity groups and those setting food policies in the City and County of Denver. So, at the project's outset we set up meetings (sometimes multiple) with our core agricultural partners (many of whom provided match for the project): the CO Potato Administrative Committee, the CO Fruit and Vegetable Growers Association, CO Wheat, the Western Horticultural Agricultural Society, and CO Beef. Accordingly, in the presentations section one can see that the project team attend each of these ag groups annual meetings each year throughout the project and held listening/discussion sessions to gain input and feedback on the project. We often included some form of data collection at these meetings – and support is demonstrated by the incredible response rates we received (see, for example, publication from Toulabi et al. 2022)!

Once strong project buy-in was established, we planned the first meeting between farmers and Denver stakeholders. For this event, we spend a significant time planning. We intentionally limited producer attendance to larger-scale (based on acreage and/or revenue) and more established growers (all but one were from multigenerational farms). We did this for three primary reasons:

- First, in the experience of the event organizers (Extension specialists, agents and researchers who collectively have decades of experience working with farmers in Colorado), small and mid-scale growers (particularly those interested in scaling up) watch what larger, more established producers do in response to new marketing opportunities. Individuals throughout rural places notice a shift in market strategy by a larger grower much more broadly than changes by small and mid-scale producers.
- Second, larger producers are generally more able and willing to take risks with new market opportunities given their enhanced ability to absorb losses and access the capital necessary to take advantage of them.
- Third, larger producers can supply products at lower prices and larger quantities relative to smaller-scale producers. Given that many of the urban market opportunities are focused on institutional procurement (e.g., jails and schools), we anticipated that the budget for procurement may be too rigid to accommodate the premium price requirements of small-scale producers.

We invited the Denver-based policymakers and buyers based on their potential to develop market opportunities that would directly align with changes to Denver's food policies. For example, three attendees represented public markets that were in the process of developing, one individual represented a company creating market opportunities for products that would

otherwise be wasted, and another was involved with procurement for the City and County of Denver. Additionally, we selected the meeting's location to signal that urban policymakers, buyers, and support organizations were committed to addressing barriers associated with connecting rural and urban areas, as many attendees had to travel ~200 miles southwest from Denver. Moreover, Gunnison, Colorado, is a midway point between the major peach, sweet corn, and potato growing regions in the state. In this way, urban policymakers were recruited based upon expressed and actual interest for encouraging regional food system economies.

The agenda and facilitation of the two-day meeting sought to build trust among participants such that their opinions and thoughts would be genuinely heard. For example, a trained facilitator led the meeting and a pre-meeting social event at a brewery was organized for the group. The research team also secured funding to cover all travel, lodging, and meal costs to lower barriers to participation.

The meeting agenda aimed to identify 'commonalities' across the various groups present. The goals of the day were outlined in a presentation that included: (1) How can Denver's food policies create new and/or higher value market opportunities for producers and regional supply chain businesses? (2) Are additional investments needed? Are there barriers to meeting the opportunities? (3) How can producers and regional leaders be more fully integrated in policymaking processes and programming priorities? We organized the agenda around speakers representing Denver policymakers and buyers in the morning, followed by producer feedback, explanations of the supply chains and matrix of factors described above, breakout sessions, and synthesis/next steps.

Following the Gunnison meeting, we continued to have meetings with key stakeholders separately, but also started work to plan the state's first Food Summit. The 2020 Colorado Food Summit (fortunately timed just prior to the shutdown caused by the pandemic) was held on January 7, 2020, in Denver. The Summit focused on creating a spotlight for the state's robust agricultural economy and building linkages with growing Denver food markets. The goal of the Colorado Food Summit was to build linkages that meet diverse urban food goals while creating viable market opportunities that builds wealth for Colorado farmers and ranchers and rural communities. The Summit sold out with 315 attendees. This was an important opportunity to build urban-rural bridges, start to dismantle some distrust, and validate some of the model assumptions.

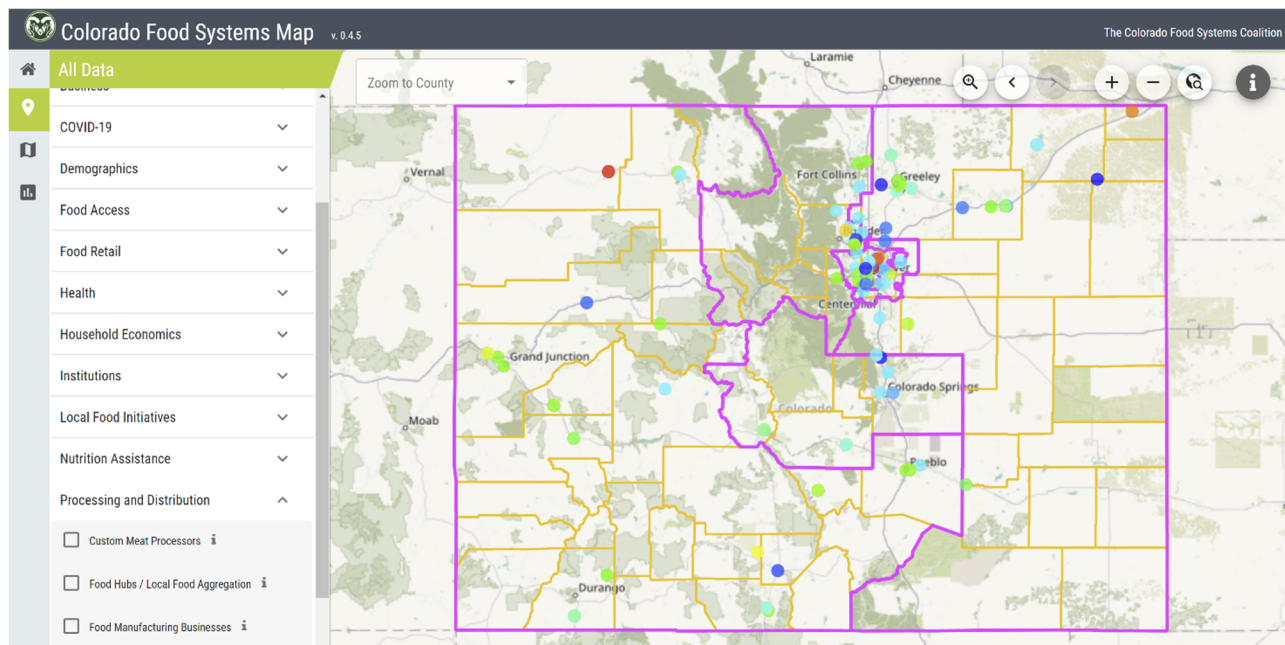
Due to COVID, we could not hold a CO Food Summit in 2021 as planned. Instead, we held 12 regional convenings around the state wherein opportunities for rural-urban connections were key themes.

A 2nd Colorado Food Summit was held on December 8, 2022. This time, it was a collaborative effort, led by CSU, the Colorado Department of Agriculture, the Colorado Food Systems Advisory Council, and all 12 of the state's local food policy councils and coalitions. The Summit inspired diverse Coloradan food and agricultural practitioners to build shared understandings of the challenges and opportunities facing our food system. In 2022, the event hosted 480 participants from Colorado (it sold out over 1 month in advance), representing all parts of the food system, including 40 youth and FFA Colorado State Officers. Complete evaluation results

from the 2022 Food Summit can be found here:

<https://foodsystems.colostate.edu/events/foodsummit2022/evaluation/> Most participants reported gaining greater knowledge about the food system, including such topics as food and agricultural policy, markets, production practices, and diverse stakeholder values and beliefs. Over 85% of survey respondents expressed that they were able to learn about the firsthand experiences of someone with a different perspective during the event. More than half (56%) of those surveyed reported finding common ground with others at the event. Perhaps most significantly, 2,125 new connections were made during the summit and more than 90% of participants were planning to follow up on connections made during the event!

Finally, we created a mapping and reporting tool to make as much data as possible publicly available, and to make it straightforward for statewide stakeholders to understand their food system. The website is available here: <https://cofoodsystemsmap.org/> During the pandemic, this tool was expanded to include key COVID-19 indicators.



2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

1. Our team conducted a rigorous multi-disciplinary literature review to ensure that we were building off of the most scientifically rigorous approaches.
2. We worked to publish articles throughout the project on sub-components that can now be cited as the full model publication goes through peer-review.
3. We made significant efforts to ensure buy in from stakeholders, including to

understand supply chain dynamics, producer-decision making, and to validate data and results.

4. Our team includes an interdisciplinary team of scholars. We spread out the budget as much as possible to support bringing many different voices into the room. We all worked together over five years to understand different disciplinary approaches and perspectives. As part of this (including internal funding from CSU), we were studied as part of a team science project the results of which were recently published in a book on food system modeling:
<https://www.sciencedirect.com/book/9780128221129/food-systems-modelling>

2.8. List key stakeholders that could be served by results of this project.

- Denver's Good Food Purchasing Program Coalition and Denver's Sustainable Food Policy Council. This includes participants from hospitals, universities, schools, farm organizations, labor, and the City of Denver, among others. The Coalition provides a space for different viewpoints to be represented while educating members about policy related to topics such as agricultural labor and animal welfare issues. Learn more: https://denvergov.org/Government/Agencies-Departments-Offices/Agencies-Departments-Offices-Directory/Public-Health-Environment/Community-Behavioral-Health/Food-System-Policies/Good-Food-Purchasing-Program?lang_update=638185934554076806#section-2
- Local/Regional/State Food Policy Councils/Coalitions. According to the Johns Hopkins University Food Policy Networks online directory, there are 322 food policy councils/coalitions in US. These Councils have grown rapidly in number over the past decade in the U.S. and across tribal nations, as networks that represent multiple stakeholders and that are either sanctioned by a government body or exist independently of government, and address food-related issues and needs within a city, county, state, tribal, multi-county or other designated region. Because collaboration amongst diverse sectors—community, government, nonprofit and private— has emerged as a long-term strategy to create systemic and meaningful improvements in the food system, the Johns Hopkins supports the Food Policy Network to build collaborative capacity of food system stakeholders to reform food systems through public policy.
- Farmers and ranchers (both throughout CO and beyond) that have the potential to sell product to schools or municipal purchasers. In Colorado, this includes are core agricultural partners: the CO Potato Administrative Committee, the CO Fruit and Vegetable Growers Association, CO Wheat, the Western Horticultural Agricultural

Society, and CO Beef.

- National Farm to School Network and their members. National Farm to School Network is an information, advocacy and networking hub for communities working to bring local food sourcing and food and agriculture education into school systems and early care and education environments. The National Farm to School Network represents more than 20,000 member organizations.
- National Sustainable Agriculture Coalition. The National Sustainable Agriculture Coalition is an alliance of grassroots organizations that advocates for federal policy reform to advance the sustainability of agriculture, food systems, natural resources, and rural communities. National Sustainable Agriculture Coalition members include more than 100 grassroots organizations.
- U.S. Department of Agriculture staff charged with program implementation; and especially employees at Agricultural Marketing Service and Food and Nutrition Service.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Diverse stakeholders were core to this project from inception. As described above, we worked very closely with these groups throughout the project, exemplified through the presentations listed, summits and convenings held, and quarterly meetings of the GFPP Coalition. We made particular efforts to spend time in rural communities and with rural stakeholders to build trust, and to bring Denver based decision-makers with us to those meetings.

Additionally, we created and regularly updated project websites and data sharing/mapping tools

- <https://cofoodsystemsmmap.org/> - this is the website that currently hosts data available for public viewing download. It is also where we will be locating the version of our model available for public use.
- We will be making all data (that isn't restricted access), the code, model inputs and outputs, the code associated with manipulating output files, the ODD (user guide), and readme file that describes the content of the GitHub site publicly available here as soon as the Nature Food article is submitted. It is already uploaded (but not yet publicly available) here: <https://github.com/CSU-Local-and-Regional-Food-Systems/foodsystemABM>
- Project website is available here: <https://foodsystems.colostate.edu/research-impacts/urban-rural-linkages/>.
- Colorado Food Summit website is here: <https://foodsystems.colostate.edu/events/colorado-food-summit/>

- 2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

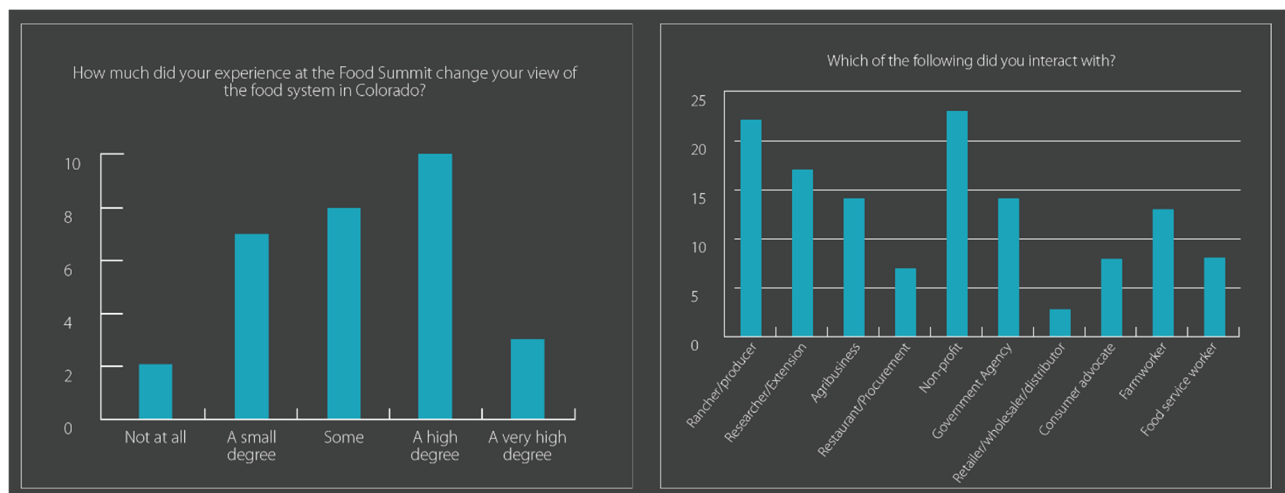
Nothing to report that isn't already described above.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Many students were involved in the project over its lifetime, including undergrad, grad and post-docs from many different disciplines. Many students had the opportunity to present at team meetings, at statewide or national conferences, to learn new methods and tools, to think interdisciplinarily, and to complete theses.

Additionally, for the first time, we held a separate youth track as part of the 2022 Colorado Food Summit. 40 youth plus the state FFA officers participated in the Summit. The Youth Summit included a series of youth-led and -focused sessions, and the opening session of the Youth Summit was livestreamed to dozens of classrooms across the state, with five rising stars in Colorado's food system taking the stage to share their projects and goals for the future. Youth participants were recruited from schools and by non-profit organizations around the state that are supporting future leaders to explore topics related to food security, regenerative farming practices, food and farm education, entrepreneurship and activism. The

Youth Summit focused on learning and building a shared understanding of the food system. Roughly 40% of Youth Summit participants reported having their views of the food system in Colorado changed to a “high degree” or greater because of their participation in the food system. Youth Summit participants reporting having interactions with actors from across the food system. One of the ways that the Summit encouraged this was through role play. Youth were asked to role play a food system stakeholder unfamiliar to them to help create a shared understanding of the challenges different food system actors face. Evaluation results show that this activity was largely successful in achieving its goal.



2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Students and post-doctoral scholars that participated in this research over the course of the project include:

- Rebecca Cleary, Agricultural and Resource Economics, Post-doctoral scholar
- Allie Bauman, Agricultural and Resource Economics, Post-doctoral scholar
- Kevin Jablonski, Forest and Rangeland Stewardship, graduate student
- James Hale, Sociology, Post-doctoral scholar
- Noereem Mena, Food Science and Human Nutrition, Postdoctoral scholar
- Hailey Summers, Mechanical Engineering, graduate student
- Javier Antoñanzas Torres, Mechanical Engineering, graduate student
- Francesco Serafin, Computer Science, postdoctoral scholar
- Stefanie Berganini, Anthropology, graduate student

- Erin Love, Agricultural and Resource Economics, graduate student
- Hailey Edmondson, Agricultural and Resource Economics, graduate student
- Pratyosh Kashyap, Agricultural and Resource Economics, graduate student
- Armen Ghazaryan, Agricultural and Resource Economics, graduate student
- Sachintha Mendis, Agricultural and Resource Economics, graduate student
- Jill Haddix, Animal Science, graduate student
- Nycole Echeverria, Ecosystem Science and Sustainability, graduate student
- Tabitha Covey, Horticulture and Landscape Architecture, graduate student
- Garrett Johnson, Mechanical Engineering, undergraduate student
- Caroline Norris, Agricultural and Resource Economics, graduate student

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.
- 3.2. Please provide a list of citations for the information products produced during the project.

In Colorado:

2018, Connecting rural Colorado communities with emerging market dynamics, state initiatives, and opportunities to think and collaborate regionally, The Colorado Trust

2018, Integrating Community and Modeling Efforts to Evaluate Impacts and Tradeoffs of Food System Interventions, Colorado Potato Administrative Committee, Fort Collins, CO

2018, Integrating Community and Modeling Efforts to Evaluate Impacts and Tradeoffs of Food System Interventions, Colorado Wheat Research Foundation, Fort Collins, CO

2018, Market opportunities, Gunnison, CO

2018, Emerging consumer opportunities for food entrepreneurs in Colorado, Denver Start Up Week, Denver, CO

2019, Assessing Market Opportunities for San Luis Valley Ag and Food Businesses, Southern Rocky Mountain Ag Conference, Monte Vista, CO

2019, Market Opportunities for Small-Scale Farmers, GoFarm, Golden, CO

2019, Emerging wholesale marketplace opportunities in Denver Metro, Colorado Fruit and Vegetable Growers Annual Meeting, Denver, CO

2019, Panel discussion on rural issues, Colorado Farm Bureau, Denver, CO

2019, Colorado Blueprint of Food and Agriculture: Engaging partners and catalyzing impacts, Colorado State University Ag Leadership Council, Denver, CO

2019, Navigating strategic food system change: A road map for community food and agriculture plans, Engagement Scholarship Consortium, Denver, CO

2019, Exploring market opportunities in the San Luis Valley: Efforts to promote resiliency and rural wealth creation, Colorado Department of Natural Resources, Denver, CO

2019, Evaluating emerging practices and markets for Colorado wheat based systems, Colorado Wheat Research Foundation, Fort Collins, CO (with M. Schipanski)

2019, Market opportunities resulting from Denver's food vision and plan: Opportunities to build rural-urban linkages, Denver Sustainable Food Policy Council, Denver, CO

2019, Leveraging urban food markets to support farmers, ranchers, and rural communities, 2019 Eastern Colorado Crop Production Conference, Fort Morgan, CO

2020, Integrating community and modeling efforts to evaluate impacts and tradeoffs of food system interventions, InTERFEWS Seminar, Colorado State University, Fort Collins, CO

2020, Exploring potential market opportunities for Colorado produce, Western Horticultural Society Annual Meeting, Grand Junction, CO

2020, CSU's Food Systems Initiative, Presentation to Centura Health's Food Security Committee, Webinar

2020, Emerging market opportunities in Denver: A 2020 update, Colorado Fruit and Vegetable Growers Association Annual Meeting, Denver, CO

2020, Food to Fork: Reviewing the COVID-19 Food Supply Chain, Denver Metro Chamber of Commerce, webinar

2020, Colorado Food Supply Chain, Cybersecurity and Infrastructure Security Agency, Department of Homeland Security, webinar

2020, Colorado State University's Task Force on Colorado Food Supply, Colorado State University Food Systems Initiative Webinar Series, webinar

2021, Colorado Proud Virtual Roundtable: Growing, Evolving and Thriving Colorado Agriculture, Colorado Department of Agriculture

2021, Cultivating relationships that work for producers and child nutrition programs, LoProCo Workshop, Nourish Colorado, virtual

2021, Consumer trends, Colorado Fruit and Vegetable Growers Association annual meeting, virtual due to COVID

2021, Market Trends Episode, Cultivation Station podcase, Colorado Department of Agriculture, virtual

2021, Conventional and alternative wheat production and supply chains, Colorado Grain School, University of Colorado- Colorado Springs, virtual

2021, Consumer food insights: COVID-19, Southern Rocky Mountain Agricultural Conference, virtual

2021, School food supply chains: Opportunities to provide support in Colorado, Session at the Office of Engagement and Extension Forum, Colorado State University

2021, Results from the Denver food acquisition survey of low-income households, Presentation to the Denver Department of Public Health and Environment, virtual

2021, Food Supply and COVID, Presentation at the Summer AgriCULTURE Institute, two presentations (Grand Junction, CO and Brighton, CO)

2021, Colorado Food Summit, Presentation to Colorado's Food Policy Councils, virtual

2021, Food Supply and COVID, COVID-19 Speaker Series, Colorado State University Alumni Association

2021, New opportunities & collaborations for Colorado's meat processing sector, KVFN radio <https://www.kvnf.org/post/new-opportunities-collaborations-colorados-meat-processing-sector#stream/0>

2021, Colorado Water Plan Scoping Workshop, Colorado Department of Natural Resources, virtual

2021, New Opportunities and Collaborations for CO's Meat Processing Sector, Colorado State University, virtual

2021, Allied Activities in the San Luis Valley to Support Community Economic Development: Opportunities to Engage, Southern Rocky Mountain Agricultural Conference, virtual

2022, Colorado Food Summit Regional Convenings (with the Colorado Food Systems Advisory Council and the Colorado Department of Agriculture) (more information: <https://foodsystems.colostate.edu/events/foodsummit2022/pre-summit/>)

- Larimer County
- Boulder County
- Yuma, CO
- Southeast, CO
- Routt and Moffat Counties, CO
- Archuleta County, CO
- Southwest, CO
- El Paso County, CO
- Denver and Jefferson Counties, CO
- Western Slope, CO
- Pueblo County, CO
- San Luis Valley, CO

2022, Integrating Community and Modeling Efforts to Evaluate Impacts and Tradeoffs of Food System Interventions, Presentation to Denver's Sustainable Food Systems Advisory Councils, Denver, CO

2022, Colorado Food Summit, Denver, CO

2023, Regional Tradeoffs Associated with Denver's Implementation of the Good Food Purchasing Program, by Scenario, Presentation to the Good Food Purchasing Program Coalition, (virtual)

U.S./International:

2017, Trends in Emerging Food Markets and Implications for Regional Economic Development, Farm to Fork Conference, Western State University, Gunnison, CO

2017, Evaluating the Rural Impacts of an Urban Local Food System Initiative, Department of Agricultural, Environmental, and Development Economics, The Ohio State University, Columbus, Ohio

2018, Eating Well Together: Developing Partnerships for a Local Food Economy, U.S. Department of Agriculture-SARE/ATTRA National Conference, St. Louis, MO

2018, Using data and new research to make the case to local and statewide decision makers for food systems investment, Harvesting Opportunities in Kansas, Lawrence KS

2018, The role of food policy councils in supporting regional food system development, Greater Kansas City Food Policy Council

2018, Integrating Community and Modeling Efforts to Evaluate Impacts and Tradeoffs of Food System Interventions, Foundation for Food and Agriculture (FFAR), Webinar

2019, Evaluating and measuring food systems impact, Massachusetts Food Systems Forum, College of the Holy Cross, Worcester, MA

2019, Innovations in the Food System: Exploring the Future of Food. 2019. Proceedings of a workshop—in brief. The National Academies of Sciences, Engineering, and Medicine.

2020, Emergency Food Provision for Children and Families during the COVID-19 Pandemic: Examples from Five U.S. Cities, Robert Wood Johnson's Healthy Eating Research Nutrition & Obesity Network, Policy Research Evaluation (HER NOPREN), virtual

2020, Leveraging urban food markets to support rural-urban linkages and rural economic development, Urban Food Systems Symposium, Kansas City, MO *invited, but the was event cancelled due to COVID-19

2020, Trends, risks, and opportunities in Western U.S. Agriculture: Implications for Extension, Western Extension Committee Organized Symposium, Western Extension Committee and Western Agricultural Economics Association annual meeting, Santa Fe, NM
*accepted but presented online due to COVID-19

2020, Putting the 'farm' back in 'farm to school': strategies to support farm viability through school markets, National Farm to Cafeteria Conference, Albuquerque, NM
*accepted but not presented due to COVID-19

2020, Food systems and sustainability. Scoping workshop on sustainable energy opportunities in food systems, Joint Institute for Strategic Energy Analysis, The National Renewable Energy Laboratory, keynote presentation, presented virtually due to COVID-19

2020, The Impacts of COVID-19 on Local Food Systems in the West, Western Agricultural Economics Association Annual Meeting, Santa Fe, NM *presented virtually due to COVID-19

2020, Price premiums for wheat attributes throughout the wheat value chain: Hedonic price analysis of flour and sandwich bread, Western Agricultural Economics Association Annual Meeting, Santa Fe, NM *presented virtually due to COVID-19

2020, Bridging Divides? A social network analysis of a statewide food summit, Agriculture and Human Values Society Annual Meeting, Athens, GA *accepted, but not presented due to COVID-19

2021, The future of sustainable food: Bridging the hunger divide, Public Affairs Conference, Missouri State University

2021, Bridging rural-urban needs, Power of Food Convening, Johns Hopkins University, virtual due to COVID

2021, Farm to school research and rural implications, U.S. Department of Agriculture, Food and Nutrition Service, virtual

2021, Food Systems and Security during COVID-19, Ethics and Ecological Economics Forum, virtual

2021, Emergency Food Provision for Children and Families during the COVID-19 Pandemic: Examples from Five U.S. Cities, Robert Wood Johnson's Healthy Eating Research Nutrition & Obesity Network, Policy Research Evaluation (HER NOPREN), virtual

2021, Emerging issues in the Child Nutrition Reauthorization Act, The Council on Food, Agricultural & Resource Economics (C-FARE) webinar

2021, COVID and Household Acquisitions: A case study from Denver, Colorado, Applied Agricultural Economics Association Annual Meeting, Austin, TX

2021, Beyond Outreach: Connecting research and Extension to support diverse stakeholders, Applied Agricultural Economics Association Annual Meeting, Austin, TX

2021, COVID and food supply chains, Applied Economic Perspectives and Policy Podcast <https://soundcloud.com/user-297451350/431-1-covid-and-food-insecurity-with-becca-jablonski-and-saweda-liverpool-tasie>

2021, Improving Public Health Impact of SNAP in the Farm Bill, Radcliffe Institute, Harvard University

2021, One Place Doesn't Fit All: Improving the Effectiveness of Sustainability Standards by Accounting for Place. Grazing in Future Multi-Scapes Virtual Conference

2022, Food Policy Structures and Governance: A Regional Food System Planning

Perspective, Virginia Tech

2022, A Systems-Based Approach to Evaluating Urban Food Procurement Policies, Department of Agricultural Economics & Rural Sociology, Auburn University (virtual)

2023, UK-US-EU opportunities for subnational food policy learning, Sustainable Food Places, London, England

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Published:

Cleary, R., A. Bonanno, A. Ghazaryan, L. Bellows, and M. McCloskey. 2021. School meals and quality of household food acquisitions. *Applied Economic Perspectives and Policy* 43: 1385-1442.

Hale, J., M. Schipanski, and M. Carolan. 2020. Just wheat transitions?: working toward constructive structural changes in wheat production. *Local Environment* 26(1):43-59. <https://doi.org/10.1080/13549839.2020.1861591>

Jablonski, B.B.R., M. Carolan, J. Hale, D. Thilmany McFadden, E. Love, L.O. Christensen, T. Covey, L. Bellows, R. Cleary, O. David, K.E. Jablonski, A. Jones, P. Meiman, J. Quinn, E. Ryan, M. Schipanski, H. Summers, and M. Uchanski. 2019. Connecting Urban Food Plans to the Countryside: Leveraging Denver's Food Vision to Explore Meaningful Rural-Urban Linkages. *Sustainability* 11(7):2022.

Jablonski, B.B.R., J. Casnovsky, J.K. Clark, R. Cleary, B. Feingold, D. Freeman, S. Gray, L. Schmitt Olabisi, X. Romeiko, C. Walsh, A.E. van den Berg. 2021. Emergency food provision for children and families during the COVID-19 pandemic: Examples from five U.S. cities. *Applied Economic Perspectives and Policy* 43(1):169-184 <https://doi.org/10.1002/aep.13096>

Jablonski, K.E., J.A. Dillon, J. Hale, B.B.R. Jablonski, and M.S. Carolan. 2020. One place doesn't fit all: Improving the effectiveness of sustainability standards by accounting for place. *Frontiers in Sustainable Food Systems*. 4:145. <https://doi.org/10.3389/fsufs.2020.557754>

Toulabi, S.B., B.B.R. Jablonski, D.G. Holm, M. Carolan, and A.L. Heuberger. 2022. Grower decision-making factors in adoption of specialty cultivators: A case study of potatoes in the San Luis Valley. *Plos One* 17(6). <https://doi.org/10.1371/journal.pone.0270636>

Under review

Love, E., B.B.R. Jablonski, D. Thilmany, and L. Bellows. A Baseline Assessment of School Food Spending and Local Procurement: Exploring the Case of Colorado's Farm to School State Incentive and Other Public Policies. Submitted to *Agricultural and Resource Economics Review*.

Wentworth, C., *M. Torres Arroyo, *R. Cavalcanti Lembi, B.J. Feingold, D. Freedman, S. Gray, B.B.R. Jablonski, K.M. Janda, P. Lemoine, A. Nielsen, X.X. Romeiko, D. Salvo, L. Schmitt Olabisi, A.E. van den Berg, *O. Yamoah. Responsive Community Engagement in Participatory Urban Food Systems Modeling: Examples from five U.S. cities. Submitted to *Environmental Science and Policy*.

In preparation

Jablonski, B.B.R., L. Bellows, *S. Berganini, A. Bonanno, R. Boone, M. Carolan, R. Cleary, O. David, J. Dillon, H. Edmondson, J. Hale, K.E. Jablonski, *P. Kashyap, *E. Love, M. McCloskey, J. Quinn, E. Ryan, M. Schipanski, F. Serafin, H. Summers, D.T. Thilmany, *J. Antoñanzas Torres, M. Uchanski, and S. Weiler. A systems-based approach to evaluating urban food procurement policies. In preparation, submitting to *Nature Food*.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Basic information about the project, including key partners and approach:

<https://foodsystems.colostate.edu/research-impacts/urban-rural-linkages/>

Information about the Colorado Food Summit:

<https://foodsystems.colostate.edu/events/foodsummit2022/>

Link to the food systems mapping, data, and reporting platform:

<https://cofoodsystems.org/mapping/>

Finally, we will be making all data (that isn't restricted access), the code, model inputs and outputs, the code associated with manipulating output files, the ODD (user guide), and readme file that describes the content of the GitHub site publicly available here as

soon as the Nature Food article is submitted.

<https://github.com/CSU-Local-and-Regional-Food-Systems/foodsystemABM>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#0	#0	
Copyrights	#0	#0	
Trademarks	#0	#0	
Inventions	#0	#0	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

N/A

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

There is more we can likely do in this space. The Nature Food article that we are in the process of finalizing includes a link to the GitHub site. In addition, with the new FFAR grant extension we are bringing together an external advisory committee (national and

international) to consider how the data and modeling work we did in CO (and will do in NY) can be more easily replicated.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

There are two places where data are made public:

<https://cofoodsystems.org/mapping/>

As mentioned above, we will be making all data (that isn't restricted access), the code, model inputs and outputs, the code associated with manipulating output files, the ODD (user guide), and readme file that describes the content of the GitHub site publicly available here as soon as the Nature Food article is submitted. It is already uploaded (but not yet publicly available) here: <https://github.com/CSU-Local-and-Regional-Food-Systems/foodsystemABM>

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

N/A

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

It will be available here shortly: <https://github.com/CSU-Local-and-Regional-Food-Systems/foodsystemABM>



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Becca B.R. Jablonski

PI Signature:  _____ Date: 5/2/2023

Authorized Signing Official (ASO) Name: Catherine Douras

ASO Signature: _____ Date: 5/11/2023